

AR-RAFI ISHRAQ

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PROFILE

Final-year Electrical and Electronic Engineering student with strong foundations in **Digital Logic Design, Digital VLSI Design, RTL Design, and Embedded Systems**. Hands-on experience with CMOS circuit design, Verilog HDL-based digital design, and hardware-oriented projects involving circuit simulation and verification. Seeking to contribute to semiconductor design.

EDUCATION

Islamic University of Technology (IUT) <i>B.Sc. in Electrical and Electronic Engineering</i> Thesis: Quad-Sector Microstructure SPR-Based PCF Sensor with Enhanced DPSS	Aug 2022 – Expected Sep 2026 CGPA: 3.94/4.00
Notre Dame College <i>Higher Secondary Certificate</i>	2019 – 2022 GPA: 5.0/5.0
Ideal School and College <i>Secondary School Certificate</i>	2017 – 2019 GPA: 5.0/5.0

ENGINEERING PROJECTS

8-Bit ALU Design and Verification <i>VLSI Design Lab — Verilog HDL — Cadence Xcelium</i> Designed an 8-bit ALU and multiple combinational and sequential circuits in Verilog HDL, and verified their functionality using testbenches and waveform analysis in Cadence Xcelium.	Aug 2026
Low-Power 6T Pass-Transistor XNOR Circuit <i>VLSI Design — PSpice — Microwind</i> Designed a low-power 6T pass-transistor XNOR circuit, verified through transient analysis, and developed its stick diagram and Microwind layout for BNN-based smart packaging.	Jul 2026
8051-Based Digital Alarm Clock Design <i>8051 Assembly — Proteus — Embedded Systems</i> Developed an 8051 microcontroller-based digital alarm clock with keypad input, LCD time display, alarm scheduling, and buzzer alert functionality using Assembly programming and Proteus simulation.	Apr 2026
Arduino-Based Automatic Fan Speed Control System <i>Control Systems — PWM Control — Hardware</i> - Developed a temperature-controlled fan speed regulation system using Arduino with sensor-based feedback and PWM-driven motor control. - Implemented automatic speed adjustment according to temperature variations for real-time thermal management.	Sep 2025
8086-Based Student Database Design <i>8086 Assembly — Microprocessor System Design</i> Developed a menu-driven student database management system using 8086 Assembly language for storing, searching, updating, and displaying student records.	Aug 2025
6-Bit SAP Computer <i>Digital Logic Design — Proteus</i> Designed and simulated a 6-bit Simple-As-Possible computer integrating registers, arithmetic circuitry, memory, datapath, and control logic to implement a functional basic computer architecture.	Nov 2023

EXTRA CURRICULAR ACTIVITIES

Project Altair – IUT Mars Rover Team <i>Electrical System Sub-Team Member</i> - Worked on circuit development, PCB design using Autodesk Eagle, wiring, system integration, testing, and troubleshooting of rover electrical systems. - Participated in the European Rover Challenge 2023 and International Rover Challenge 2024 as part of Project Altair. The team received the Best Science Team award at IRC 2024.	2023 – 2025
IUT Robotics Society – IUT RS - Organized flagship events including TECHATHON 1.0 2024 and Drone Workshop 2024, overseeing planning, coordination, and execution. - Contributed to the design of competition arenas for Soccerbot, Line Following Robot (LFR), and RC Bot.	2023 – 2025

INDUSTRIAL TRAINING

Ghorashal Training Center / Power Station <i>Industrial Training</i> • Gained exposure to power-station operations, electrical systems, and power-generation processes.	Oct 2025
Bangladesh Atomic Energy Commission (BAEC) <i>Industrial Training</i> • Witnessed solar-cell fabrication and visited the 3 MW TRIGA Mark-II research reactor.	Oct 2025
Robi Axiata Limited – Head Office <i>Industrial Visit</i> • Learned about telecom operations, technical workflows, and service management.	Oct 2025
Walton Hi-Tech Industries PLC <i>Industrial Visit</i> • Gained exposure to industrial manufacturing, quality control, and automation processes.	Oct 2025

TECHNICAL SKILLS

RTL & Digital Design: Verilog HDL, SystemVerilog, RTL Design, combinational and sequential logic design, test-bench development, functional verification, waveform analysis, RTL synthesis

Cadence EDA Tools: Xcelium, Genus, Virtuoso

Circuit & System Simulation: PSpice, OrCAD, Proteus, PSIM, MATLAB/Simulink

Programming & Assembly: C, Python, MATLAB, 8051 Assembly, 8086 Assembly

LEADERSHIP EXPERIENCE

Project Altair – IUT Mars Rover Team • Senior Member (Electrical Sub-Team) • Junior Member (Electrical Sub-Team)	2024 – 2025 2023 – 2024
IUT Robotics Society • Assistant Head (Event & Workshop) • Sub Executive Officer	2024 – 2025 2023 – 2024

ADDITIONAL INFORMATION

Languages: Bangla (Native), English (Fluent)

REFERENCES

Dr. Mohammad Rakibul Islam Professor, Department of EEE Islamic University of Technology (IUT) rakibultowhid@yahoo.com +8802 99669 1254–59	Dr. Md. Masum Billah Assistant Professor, Department of EEE Islamic University of Technology (IUT) masum.billah@iut-dhaka.edu +8801748806097
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